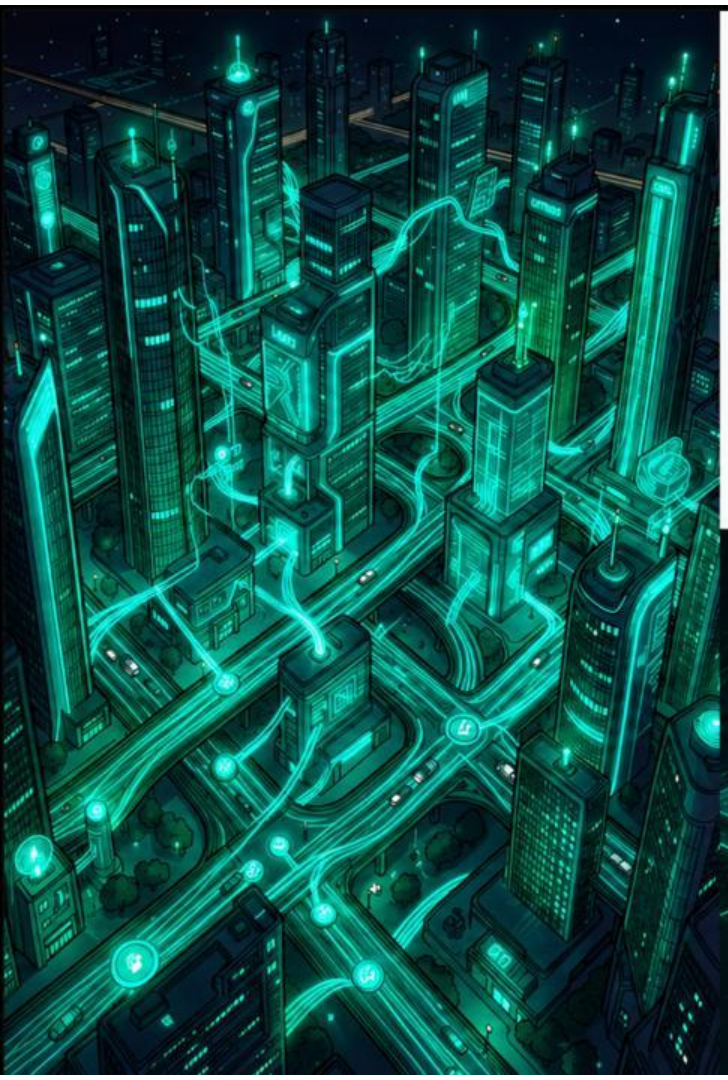


UrbanFlow

Smart Parking Management System



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RFID Access

Real-time Monitoring

Automated Gate Control

SUSTAINABLE.
EFFICIENT. CONNECTED.

Problem Statement



No Real-Time Parking Visibility



Time Wasted Searching for Parking



Traffic Congestion



Fuel & Energy Wastage



No Advance Reservation System



Underutilized Parking Infrastructure

THE PROBLEM

Parking Remains a Major Challenge

⏱️ Time Lost

In dense urban areas, drivers spend an average of **17 minutes** searching for parking.

⛽ Fuel Wasted

Up to **30% of urban traffic** comes from vehicles circling in search of available spaces.

🚙 Increased Congestion

Poor parking management contributes to bottlenecks, accidents, and commuter frustration.

🔒 Limited Security

Manual systems lack authentication, audit trails, and real-time visibility.



OUR SOLUTION

Introducing UrbanFlow

A fully automated, IoT-enabled smart parking system that eliminates guesswork – from entry to exit.

RFID Authentication

Instant vehicle identification – no tickets, no queues.

Real-Time Slot Monitoring

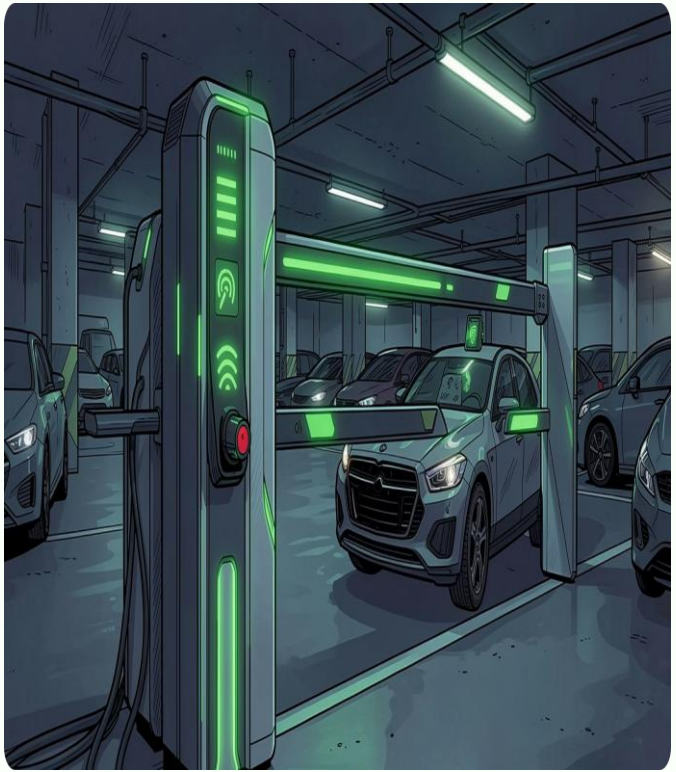
IR sensors track every slot, updating the dashboard live.

Automated Gate Control

Servo motor opens and closes gates on authorized access only.

Smart Dashboard

Live slot count, alerts, and analytics in one clean interface.

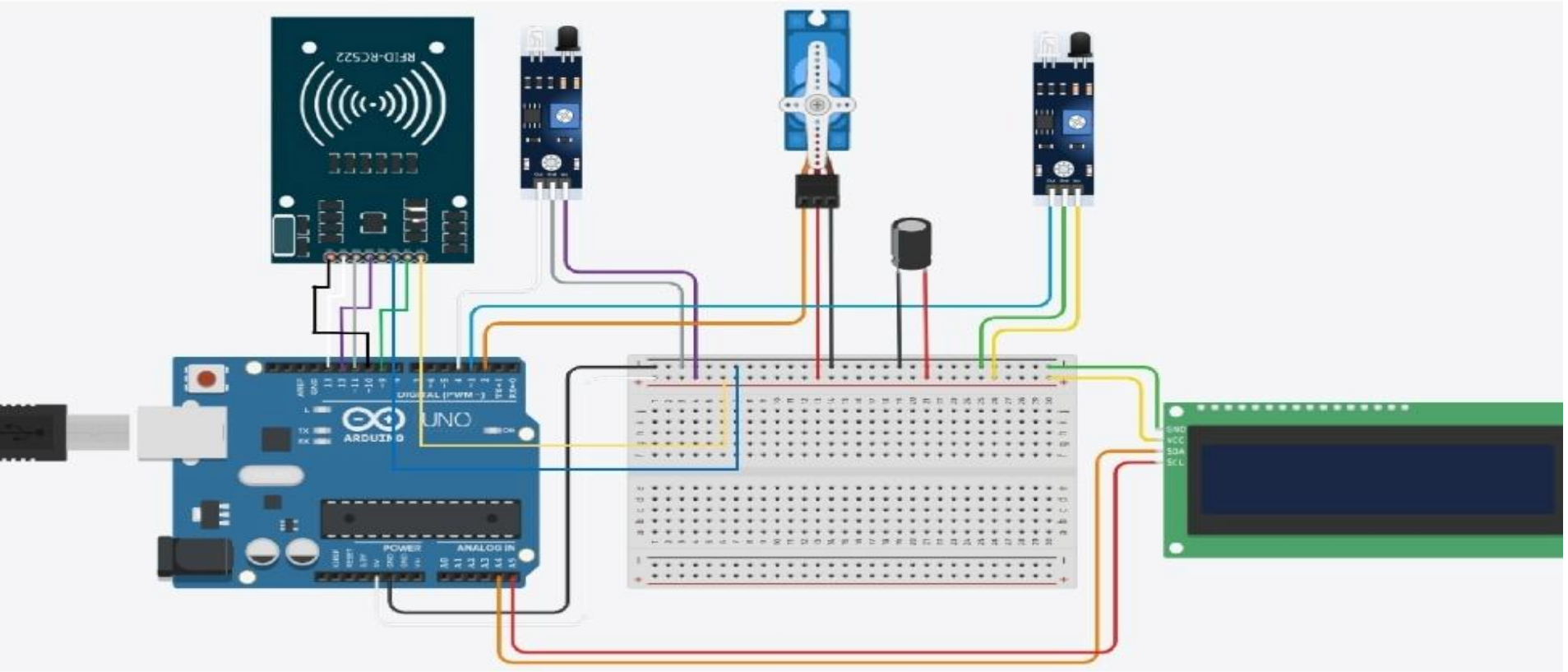


UrbanFlow

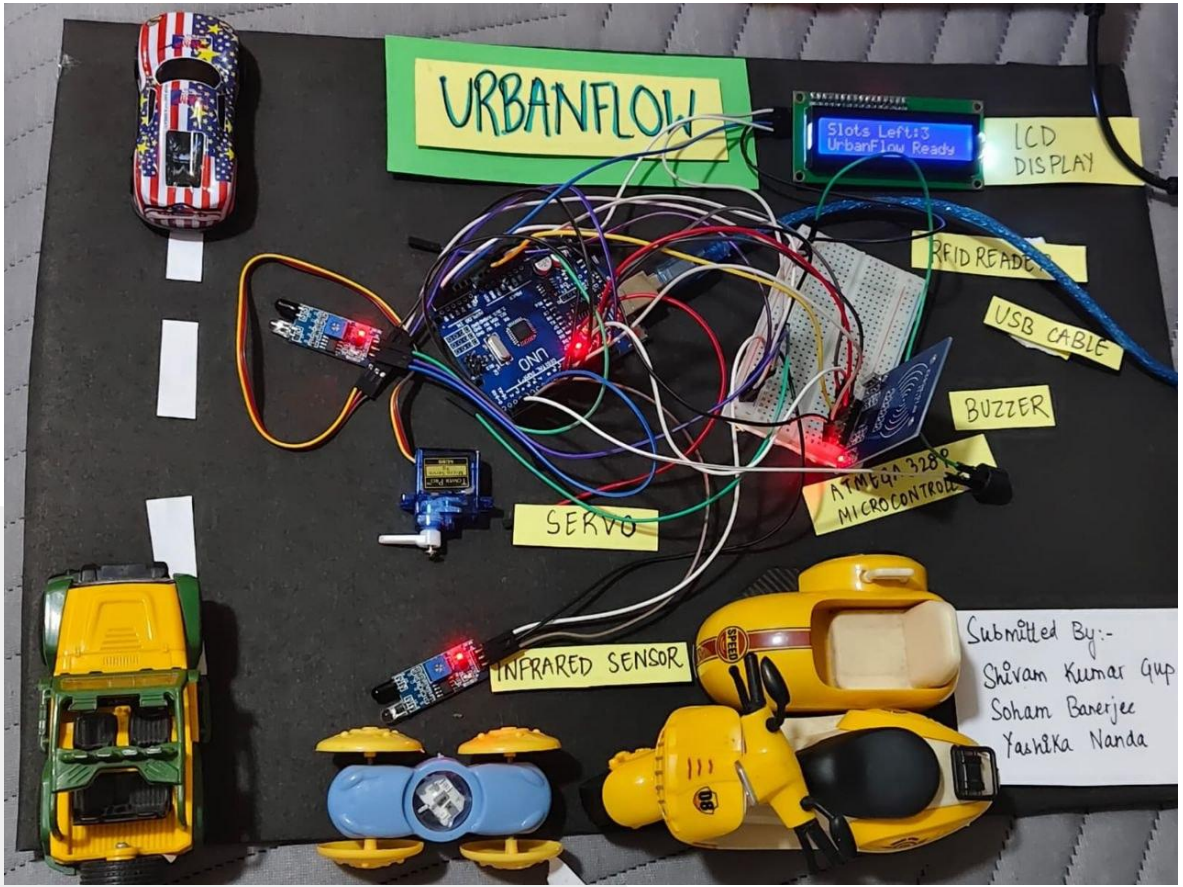
SOFTWARE WORKFLOW



INTELLIGENT IOT HARDWARE ARCHITECTURE



Prototype





Arduino Uno



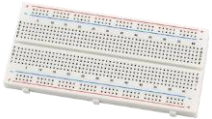
RFID Module



IR Sensor



Servo Motor



Breadboard



16*2 LCD Display



Jumper Wire

LIVE DEMO

Prototype in Action

RFID UID Detection

Instant recognition of registered vehicles – unauthorized tags are denied entry.

Live Slot Count

Dashboard updates in real time as vehicles enter and exit the facility.

Parking Full Alert

System auto-blocks entry and displays "FULL" on LCD when capacity is reached.

Automated Gate

Servo motor responds instantly – no human intervention required.

CHALLENGES & SOLUTIONS

Challenges We Faced

Developing UrbanFlow presented several technical hurdles, which we addressed through robust engineering and iterative problem-solving.

Hardware Integration

Connecting diverse components (RFID, IR, Arduino) required a modular approach with standardized communication protocols. Extensive testing ensured seamless interaction and data flow.

Real-Time Data Sync

Maintaining sub-second latency for sensor data and gate control was critical. We implemented WebSockets for instant, bidirectional communication and optimized data structures for efficient processing.

Sensor Accuracy

Ensuring reliable occupancy detection from IR sensors in varying conditions posed a challenge. We developed a multi-point calibration routine and integrated software filters to minimize false positives and negatives.

Software Optimization

Optimizing the backend for rapid response and scalability was key. We utilized Python's FastAPI for its asynchronous capabilities and fine-tuned database queries in SQLite to handle high transaction volumes efficiently.

WHAT'S NEXT

Future Scope & Impact

Expanding from a smart parking system into a connected urban platform that improves mobility, efficiency, and experience at scale.

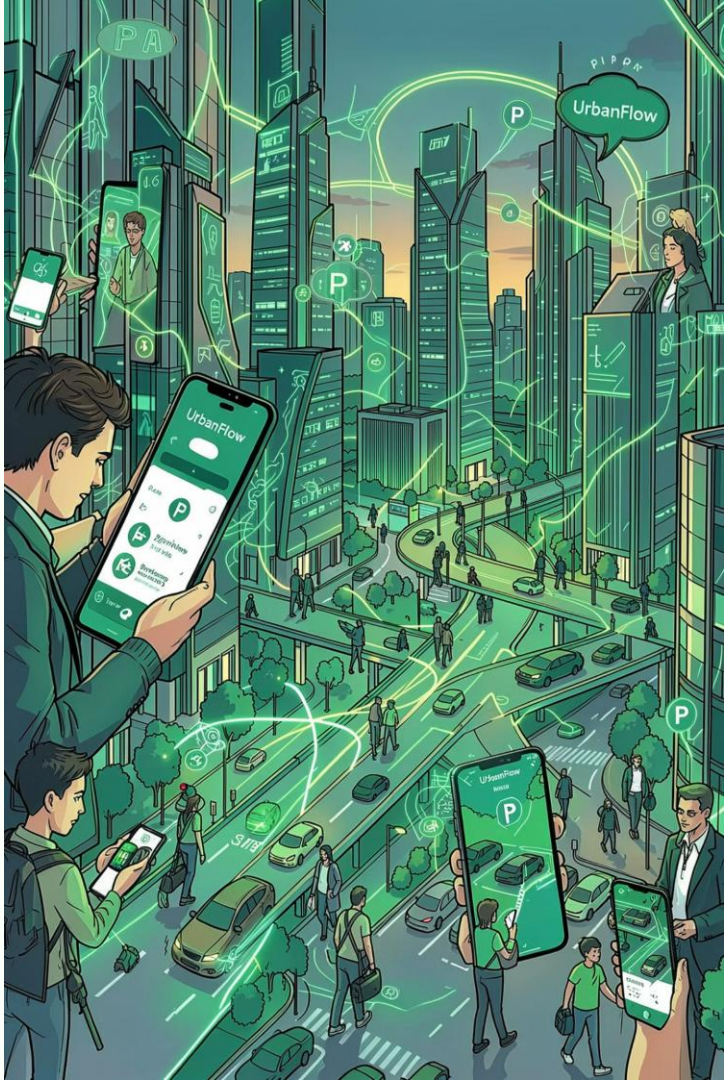
1. **Mobile App**
   Enable slot reservation and live navigation.
2. **Online Payments**
   Streamline exit with integrated checkout.
3. **AI Analytics**
   Use predictive insights for occupancy and demand forecasting.
4. **5V Power Supply**
   Support larger deployments with a dedicated power supply unit.

🌍 Deployment Impact

Smart Cities
Power city-wide parking intelligence networks.

Hospitals & Malls
Improve mobility and management across high-traffic facilities.

Colleges & Campuses
Deliver secure, student-friendly access experiences.





THANK YOU